The Transfer-Matrix Method for Planar Multilayer Systems

A Step-by-Step Derivation

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The transfer-matrix method (TMM) provides a compact framework for computing the optical response of planar multilayer systems. This document derives the reflection and transmission amplitudes for a stack of homogeneous, isotropic, possibly absorbing layers, serving as the physical foundation of the `tmm-core` Python package. The derivation proceeds in two stages: the single-interface Fresnel coefficients are first cast in the compact form used throughout, and the full transfer matrix is then assembled from two elementary building blocks—a propagation matrix for each layer and an interface matrix for each boundary—from which the stack amplitudes follow via a single 2×2 linear system. The conventions of Lekner [1] and Yeh [2] are adopted throughout; the branch-cut treatment of complex wavenumbers and the transfer-matrix organization follow Byrnes [3]. The time convention is $e^{-\mathrm{i}\omega t}$, so that a forward-propagating plane wave carries the spatial factor $e^{+\mathrm{i}q_j z}$ and the imaginary part of the refractive index is non-negative for passive (absorbing) media.

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I. SETUP

Consider a stratified medium whose refractive index depends only on the coordinate z normal to the interfaces and is piecewise constant inside each layer. Light propagates in the xz-plane, and the interfaces lie at $z = \text{const}$. Each layer j is characterized by a complex refractive index

$$N_j = n_j + \mathrm{i}\kappa_j, \qquad \kappa_j \geq 0. \tag{1}$$

With the time convention $e^{-\mathrm{i}\omega t}$, $\kappa_j \geq 0$ ensures the layer is passive (absorbing rather than amplifying).

a. Labeling. The stack consists of N finite layers numbered $j = 1, 2, \ldots, N$, bounded by two semi-infinite half-spaces labeled $j = 0$ (incident medium) and $j = N + 1$ (substrate). The j-th interface lies at $z = z_j$, and layer j has thickness $d_j = z_j - z_{j-1}$.

II. WAVENUMBERS

The vacuum wavenumber corresponding to wavelength λ_0 is

$$k_0 \;=\; \frac{2\pi}{\lambda_0}. \tag{2}$$

Inside layer j the plane-wave dispersion relation gives the complex wavenumber

$$k_j \;=\; N_j\, k_0 \;\in\; \mathbb{C}. \tag{3}$$

Translation invariance in x forces the tangential wavevector component to be identical in every layer:

$$k_x \;=\; k_0 \sin\theta_0 \;=\; k_j \sin\theta_j \;=\; \text{const}. \tag{4}$$

Here θ_0 is the (real) angle of incidence in the incident half-space and θ_j is the generally complex refraction angle in layer j. Dividing by k_0 recovers Snell's law, $n_0 \sin\theta_0 = N_j \sin\theta_j$.

The normal wavevector component in layer j follows from $|\mathbf{k}_j|^2 = k_j^2$,

$$k_z^{(j)} \;=\; \sqrt{k_j^2 - k_x^2}\,. \tag{5}$$

The square root in (5) has two branches. The physical branch is fixed by two requirements:

1. A forward-propagating wave entering an absorbing medium must decay, not grow, with depth.

2. Above the critical angle the transmitted wave is evanescent and must decay away from the interface.

Both requirements are captured by the branch-cut convention of Byrnes [3]:

$$\mathrm{Im}\big[k_z^{(j)}\big] \;\geq\; 0, \qquad \text{and if } \mathrm{Im}\big[k_z^{(j)}\big] = 0, \text{ then } \mathrm{Re}\big[k_z^{(j)}\big] > 0. \tag{6}$$

This convention must be enforced explicitly in code: the principal-branch square root of standard numerical libraries (e.g. `numpy.sqrt`, which returns $\mathrm{Re} \geq 0$) does not generally satisfy (6).

Equivalently, $k_z^{(j)} = k_j \cos\theta_j$, consistent with (5): substituting $k_j \sin\theta_j = k_x$ from (4),

$$k_j \cos\theta_j \;=\; \sqrt{k_j^2 - k_j^2 \sin^2\theta_j} \;=\; \sqrt{k_j^2 - k_x^2}\,. \tag{7}$$

It is convenient to define

$$q_j \;\equiv\; k_z^{(j)} \;=\; \sqrt{(N_j k_0)^2 - \beta^2}\,, \qquad Q_j \;\equiv\; \frac{q_j}{N_j^2}, \qquad \beta \;\equiv\; k_x. \tag{8}$$

The quantity q_j is the natural variable for s-polarization; Q_j for p-polarization [1].

III. FRESNEL COEFFICIENTS AT A SINGLE INTERFACE

The Fresnel reflection coefficients in the notation of [1, 2] are

$$r_s^{(j,j+1)} \;=\; \frac{k_z^{(j)} - k_z^{(j+1)}}{k_z^{(j)} + k_z^{(j+1)}}, \tag{9}$$

$$r_p^{(j,j+1)} \;=\; \frac{N_j^2\, k_z^{(j+1)} \;-\; N_{j+1}^2\, k_z^{(j)}}{N_{j+1}^2\, k_z^{(j)} \;+\; N_j^2\, k_z^{(j+1)}}. \tag{10}$$

With the abbreviations (8), these take a compact symmetric form:

$$r_s^{(j,j+1)} \;=\; \frac{q_j - q_{j+1}}{q_j + q_{j+1}}, \tag{11}$$

$$r_p^{(j,j+1)} \;=\; -\,\frac{Q_j - Q_{j+1}}{Q_j + Q_{j+1}}. \tag{12}$$

At normal incidence, $q_j = N_j k_0$ and $Q_j = k_0/N_j$, so

$$r_s(0) \;=\; \frac{N_j - N_{j+1}}{N_j + N_{j+1}}, \qquad r_p(0) \;=\; -\,\frac{1/N_j - 1/N_{j+1}}{1/N_j + 1/N_{j+1}} \;=\; \frac{N_j - N_{j+1}}{N_j + N_{j+1}}. \tag{13}$$

At normal incidence the two polarizations are physically indistinguishable, and the form (12) is chosen precisely so that $r_s(0) = r_p(0)$ without a sign flip. This is the convention of Lekner [1]

and Yeh [2], corresponding to "Convention A" in Byrnes [3, Appendix A]. Some sources (including Wikipedia) use the opposite sign, $r_s(0) = -r_p(0)$. Both conventions yield the same reflectance $R_p = |r_p|^2$ but differ in the phase of r_p and hence in ellipsometric quantities such as $\Delta = \arg(-r_p/r_s)$.

With the same conventions, the transmission coefficients are [1]

$$t_s^{(j,j+1)} = \frac{2q_j}{q_j + q_{j+1}}, \tag{14}$$

$$t_p^{(j,j+1)} = \frac{N_j}{N_{j+1}} \frac{2Q_j}{Q_j + Q_{j+1}}. \tag{15}$$

IV. CONSISTENCY RELATIONS

Direct algebraic manipulation of (11)–(15) yields a set of identities that serve both as checks and as the algebraic input for the transfer-matrix derivation below.

$$1 + r_s^{(j,j+1)} = t_s^{(j,j+1)}. \tag{16}$$

Derivation.

$$1 + \frac{q_j - q_{j+1}}{q_j + q_{j+1}} = \frac{(q_j + q_{j+1}) + (q_j - q_{j+1})}{q_j + q_{j+1}} = \frac{2q_j}{q_j + q_{j+1}} = t_s^{(j,j+1)}. \tag{17}$$

The p-polarization analogue is

$$1 - r_p^{(j,j+1)} = \frac{N_{j+1}}{N_j} t_p^{(j,j+1)}. \tag{18}$$

Derivation. Using (12),

$$1 - r_p^{(j,j+1)} = 1 + \frac{Q_j - Q_{j+1}}{Q_j + Q_{j+1}} = \frac{2Q_j}{Q_j + Q_{j+1}}. \tag{19}$$

Comparing with (15), one has $\frac{N_{j+1}}{N_j} t_p^{(j,j+1)} = \frac{2Q_j}{Q_j+Q_{j+1}}$, confirming (18).

For both polarizations, the reflection coefficient is antisymmetric under interchange of the two media,

$$r^{(j+1,j)} = -r^{(j,j+1)}, \tag{20}$$

as is evident by inspection of (11) and (12). Combined with the transmission coefficients, this yields the Stokes identity

$$t^{(j,j+1)} t^{(j+1,j)} = 1 - \left(r^{(j,j+1)}\right)^2. \tag{21}$$

Derivation (s-polarization). From (14),

$$t_s^{(j,j+1)}\, t_s^{(j+1,j)} \;=\; \frac{2q_j}{q_j+q_{j+1}} \cdot \frac{2q_{j+1}}{q_j+q_{j+1}} \;=\; \frac{4\,q_j\,q_{j+1}}{(q_j+q_{j+1})^2}, \tag{22}$$

while

$$1-\left(r_s^{(j,j+1)}\right)^2 \;=\; \frac{(q_j+q_{j+1})^2-(q_j-q_{j+1})^2}{(q_j+q_{j+1})^2} \;=\; \frac{4\,q_j\,q_{j+1}}{(q_j+q_{j+1})^2}. \tag{23}$$

The p-polarization case follows the same pattern with $q_j \to Q_j$; the prefactors from (15) cancel as $(N_j/N_{j+1})(N_{j+1}/N_j) = 1$.

The Stokes identity (21) is the key algebraic input for the interface matrix derived in Sec. V C.

V. TRANSFER-MATRIX FORMALISM

The goal of this section is to reduce the physics of the entire stack to a single 2×2 matrix M relating the field amplitudes in the two outer half-spaces. The strategy is to decompose the stack into two types of elementary building blocks:

- Inside layer j, the field is a superposition of a forward and a backward plane wave. Propagation across the layer rescales the two amplitudes by opposite phases; this gives the *propagation matrix* P_j (Sec. V B).
- At interface $z = z_j$, the amplitudes on either side are linked by Fresnel coefficients; this gives the *interface matrix* $T^{(j,j+1)}$ (Sec. V C).

The full transfer matrix is the ordered product of these building blocks (Sec. V D), and the reflection and transmission amplitudes follow from the single 2×2 system (45).

A. Field representation

Inside layer j, between the interfaces at z_{j-1} and z_j (thickness $d_j = z_j - z_{j-1}$), the field satisfies a one-dimensional Helmholtz equation whose general solution is

$$E_j(z) \;=\; A_j\, e^{+\mathrm{i}q_j z} \;+\; B_j\, e^{-\mathrm{i}q_j z}. \tag{24}$$

With the convention $e^{-\mathrm{i}\omega t}$, the first term propagates in the $+z$ direction and the second in the $-z$ direction. The complex amplitudes A_j and B_j are constant throughout the layer, but their relative phase depends on the position at which (24) is evaluated.

To handle the phase bookkeeping systematically, we define amplitudes at the left edge ($z = z_{j-1}$) and right edge ($z = z_j$) of the layer:

$$\alpha_j^{(L)} \equiv A_j\, e^{+\mathrm{i}q_j z_{j-1}}, \qquad \beta_j^{(L)} \equiv B_j\, e^{-\mathrm{i}q_j z_{j-1}}, \tag{25}$$

$$\alpha_j^{(R)} \equiv A_j\, e^{+\mathrm{i}q_j z_j}, \qquad \beta_j^{(R)} \equiv B_j\, e^{-\mathrm{i}q_j z_j}. \tag{26}$$

Here α always denotes the forward-wave amplitude and β the backward-wave amplitude; the superscript indicates which edge of the layer is used.

Defining the layer phase

$$\phi_j \equiv q_j\, d_j = q_j(z_j - z_{j-1}), \tag{27}$$

equations (25)–(26) give the left-to-right relations

$$\alpha_j^{(R)} = \alpha_j^{(L)}\, e^{+\mathrm{i}\phi_j}, \qquad \beta_j^{(R)} = \beta_j^{(L)}\, e^{-\mathrm{i}\phi_j}. \tag{28}$$

The forward wave accumulates phase $+\phi_j$ crossing from left to right; the backward wave accumulates $-\phi_j$ in the opposite direction. For an absorbing layer, $\mathrm{Im}(\phi_j) > 0$, so $|e^{-\mathrm{i}\phi_j}| < 1$: the backward wave is smaller at the right edge than at the left edge, as expected for a wave physically traveling right to left.

B. Propagation matrix

a. Direction convention. The total matrix M maps substrate-side amplitudes to incident-side amplitudes, i.e., *right to left*. Every building block must follow the same convention. Inverting (28),

$$\alpha_j^{(L)} = \alpha_j^{(R)}\, e^{-\mathrm{i}\phi_j}, \qquad \beta_j^{(L)} = \beta_j^{(R)}\, e^{+\mathrm{i}\phi_j}, \tag{29}$$

which in matrix form reads

$$\boxed{\begin{pmatrix} \alpha_j^{(L)} \\ \beta_j^{(L)} \end{pmatrix} = \underbrace{\begin{pmatrix} e^{-\mathrm{i}\phi_j} & 0 \\ 0 & e^{+\mathrm{i}\phi_j} \end{pmatrix}}_{P_j} \begin{pmatrix} \alpha_j^{(R)} \\ \beta_j^{(R)} \end{pmatrix}} \tag{30}$$

C. Interface matrix

At the interface $z = z_j$ separating the right edge of layer j (amplitudes $\alpha_j^{(R)}, \beta_j^{(R)}$) from the left edge of layer $j{+}1$ (amplitudes $\alpha_{j+1}^{(L)}, \beta_{j+1}^{(L)}$), the amplitudes on either side are linked by the local Fresnel coefficients via two physical processes:

- the forward wave in $j{+}1$ receives contributions from (i) the transmitted forward wave from j and (ii) the reflected backward wave from $j{+}1$;

- the backward wave in j receives contributions from (i) the reflected forward wave from j and (ii) the transmitted backward wave from $j{+}1$.

In equations,

$$\alpha_{j+1}^{(L)} \;=\; t^{(j,j+1)}\,\alpha_j^{(R)} \;+\; r^{(j+1,j)}\,\beta_{j+1}^{(L)}, \tag{$\star$}$$

$$\beta_j^{(R)} \;=\; r^{(j,j+1)}\,\alpha_j^{(R)} \;+\; t^{(j+1,j)}\,\beta_{j+1}^{(L)}. \tag{$\star\star$}$$

The right-to-left convention requires expressing the right-edge amplitudes of layer j in terms of the left-edge amplitudes of layer $j{+}1$.

a. Step 1: solve ($\star$) *for* $\alpha_j^{(R)}$. Using the reversal identity (20), $r^{(j+1,j)} = -r^{(j,j+1)}$,

$$t^{(j,j+1)}\,\alpha_j^{(R)} \;=\; \alpha_{j+1}^{(L)} - r^{(j+1,j)}\,\beta_{j+1}^{(L)} \;=\; \alpha_{j+1}^{(L)} + r^{(j,j+1)}\,\beta_{j+1}^{(L)}, \tag{31}$$

so

$$\alpha_j^{(R)} \;=\; \frac{1}{t^{(j,j+1)}}\Big(\alpha_{j+1}^{(L)} + r^{(j,j+1)}\,\beta_{j+1}^{(L)}\Big). \tag{32}$$

b. Step 2: substitute into ($\star\star$). Inserting (32) into ($\star\star$),

$$\beta_j^{(R)} \;=\; \frac{r^{(j,j+1)}}{t^{(j,j+1)}}\,\alpha_{j+1}^{(L)} \;+\; \underbrace{\left(\frac{\big(r^{(j,j+1)}\big)^2}{t^{(j,j+1)}} \;+\; t^{(j+1,j)}\right)}_{\text{to be simplified}}\beta_{j+1}^{(L)}. \tag{33}$$

c. Step 3: Stokes collapse. The parenthesized expression in (33) reduces to $1/t^{(j,j+1)}$ by the Stokes identity (21):

$$\frac{\big(r^{(j,j+1)}\big)^2}{t^{(j,j+1)}} + t^{(j+1,j)} \;=\; \frac{\big(r^{(j,j+1)}\big)^2 + t^{(j,j+1)}\,t^{(j+1,j)}}{t^{(j,j+1)}} \;=\; \frac{\big(r^{(j,j+1)}\big)^2 + 1 - \big(r^{(j,j+1)}\big)^2}{t^{(j,j+1)}} \;=\; \frac{1}{t^{(j,j+1)}}. \tag{34}$$

Hence

$$\beta_j^{(R)} \;=\; \frac{1}{t^{(j,j+1)}}\Big(r^{(j,j+1)}\,\alpha_{j+1}^{(L)} + \beta_{j+1}^{(L)}\Big). \tag{35}$$

This is the step that makes the whole construction work: without the Stokes identity, the interface matrix would not take a simple closed form.

Combining (32) and (35),

$$\boxed{\begin{pmatrix} \alpha_j^{(R)} \\ \beta_j^{(R)} \end{pmatrix} = \underbrace{\frac{1}{t^{(j,j+1)}} \begin{pmatrix} 1 & r^{(j,j+1)} \\ r^{(j,j+1)} & 1 \end{pmatrix}}_{T^{(j,j+1)}} \begin{pmatrix} \alpha_{j+1}^{(L)} \\ \beta_{j+1}^{(L)} \end{pmatrix}} \tag{36}$$

Like P_j, the interface matrix $T^{(j,j+1)}$ maps right-edge amplitudes to left-edge amplitudes.

D. Total transfer matrix and optical amplitudes

Both building blocks P_j and $T^{(j,j+1)}$ map right-side amplitudes to left-side amplitudes. Chaining them from the substrate inward produces a matrix that carries amplitudes at the substrate to amplitudes in the incident medium. Starting from the substrate amplitudes (A_{N+1}, B_{N+1}) and prepending one factor at a time:

$$\begin{pmatrix} \alpha_N^{(R)} \\ \beta_N^{(R)} \end{pmatrix} = T^{(N,N+1)} \begin{pmatrix} A_{N+1} \\ B_{N+1} \end{pmatrix}, \tag{37}$$

$$\begin{pmatrix} \alpha_N^{(L)} \\ \beta_N^{(L)} \end{pmatrix} = P_N \, T^{(N,N+1)} \begin{pmatrix} A_{N+1} \\ B_{N+1} \end{pmatrix}, \tag{38}$$

$$\begin{pmatrix} \alpha_{N-1}^{(R)} \\ \beta_{N-1}^{(R)} \end{pmatrix} = T^{(N-1,N)} \, P_N \, T^{(N,N+1)} \begin{pmatrix} A_{N+1} \\ B_{N+1} \end{pmatrix}, \tag{39}$$

$$\vdots \tag{40}$$

After N propagation steps and $N+1$ interface steps, one arrives at the amplitudes in the incident half-space. The two semi-infinite half-spaces carry no finite thickness, so no propagation matrix appears for them:

$$\begin{pmatrix} A_0 \\ B_0 \end{pmatrix} = T^{(0,1)} \, P_1 \, T^{(1,2)} \, P_2 \cdots P_N \, T^{(N,N+1)} \begin{pmatrix} A_{N+1} \\ B_{N+1} \end{pmatrix}. \tag{41}$$

Grouping each propagation matrix with the interface matrix immediately to its right gives the compact form

$$\boxed{M = T^{(0,1)} \prod_{j=1}^{N} \left(P_j \, T^{(j,j+1)} \right)} \tag{42}$$

where the product is ordered left to right with increasing j.

a. Remark: relation to Byrnes. Byrnes [3, Eq. (11)] combines P_n and $T^{(n,n+1)}$ into a single matrix $M_n = P_n T^{(n,n+1)}$ and writes the stack product in terms of M_n factors with an outer $T^{(0,1)}$. The present form (42) is mathematically identical but keeps the propagation and interface contributions explicitly separate.

The physical boundary conditions are a unit-amplitude incident wave with no wave incoming from the substrate:

$$A_0 = 1, \qquad B_0 = r, \qquad A_{N+1} = t, \qquad B_{N+1} = 0, \tag{43}$$

where r and t are the reflection and transmission amplitudes of the entire stack. Substituting (43) into (42),

$$\begin{pmatrix} 1 \\ r \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} t \\ 0 \end{pmatrix} = \begin{pmatrix} M_{11}\, t \\ M_{21}\, t \end{pmatrix}. \tag{44}$$

The first row gives $t = 1/M_{11}$ and the second gives $r = M_{21}/M_{11}$:

$$\boxed{t \;=\; \frac{1}{M_{11}}, \qquad r \;=\; \frac{M_{21}}{M_{11}}} \tag{45}$$

This is the main result of the transfer-matrix formalism. The reflectance follows as $R = |r|^2$.

VI. ABSORPTION PER LAYER

With r and t determined, the field inside the stack is fully fixed, and the time-averaged power absorbed in each finite layer follows from energy conservation: it equals the Poynting flux entering the layer at its left edge minus the flux leaving at its right edge. All quantities below are normalized to the incident time-averaged power, so the dimensionless flux $\mathcal{S}(z)$ is the forward power at position z relative to the incoming power far from the stack.

Using the left-edge amplitudes $(\alpha_j^{(L)}, \beta_j^{(L)})$ from (25), the forward and backward field components at any $z \in [z_{j-1}, z_j]$ are

$$E_f^{(j)}(z) \;=\; \alpha_j^{(L)}\, e^{+\mathrm{i}q_j\,(z-z_{j-1})}, \qquad E_b^{(j)}(z) \;=\; \beta_j^{(L)}\, e^{-\mathrm{i}q_j\,(z-z_{j-1})}. \tag{46}$$

At $z = z_j$ these reduce to $\alpha_j^{(R)}$ and $\beta_j^{(R)}$ by (28).

Evaluating $\frac{1}{2}\,\mathrm{Re}[\hat{\mathbf{z}} \cdot (\mathbf{E}^* \times \mathbf{H})]$ with the explicit field expressions of Byrnes [3, Eqs. (4)–(5)] and normalizing by the incident power gives the normalized Poynting flux

$$\mathcal{S}_s^{(j)}(z) \;=\; \frac{1}{\mathrm{Re}[q_0]}\, \mathrm{Re}\big[\, q_j \big(E_f^* + E_b^*\big)\big(E_f - E_b\big)\big]\,, \tag{47}$$

$$\mathcal{S}_p^{(j)}(z) \;=\; \frac{1}{\mathrm{Re}[\bar{q}_0]}\, \mathrm{Re}\big[\, \bar{q}_j \big(E_f - E_b\big)\big(E_f^* + E_b^*\big)\big]\,, \tag{48}$$

where E_f, E_b abbreviate $E_f^{(j)}(z), E_b^{(j)}(z)$ and

$$\bar{q}_j \;\equiv\; k_0\, N_j \cos\theta_j^* \;=\; \frac{N_j}{N_j^*}\, q_j^* \;=\; |N_j|^2\, Q_j^*. \tag{49}$$

For s-polarization the natural prefactor is q_j; the p-polarization convention adopted here ($r_s = r_p$ at normal incidence, i.e. the E-field-based choice, opposite to the H-field-based Convention A of Byrnes [3]) produces instead the conjugated cosine combination $N_j \cos\theta_j^*$, which equals $\bar{q}_j/k_0$. The factor k_0 cancels against the denominator. The same sign convention also swaps the roles of $(E_f \pm E_b)$ in the p-Poynting scalar product relative to Byrnes: here the combination $(E_f - E_b)(E_f^* + E_b^*)$ appears, so that (47) and (48) are structurally identical, differing only in the replacement $q_j \to \bar{q}_j$.

Although equations (47) and (48) are used directly in the code, it is instructive to expand the scalar products into a diagonal and an interference term:

$$(E_f^* + E_b^*)(E_f - E_b) \;=\; \big(|E_f|^2 - |E_b|^2\big) \;+\; 2\mathrm{i}\,\mathrm{Im}\big(E_f E_b^*\big), \tag{50}$$

$$(E_f - E_b)(E_f^* + E_b^*) \;=\; \big(|E_f|^2 - |E_b|^2\big) \;+\; 2\mathrm{i}\,\mathrm{Im}\big(E_f E_b^*\big). \tag{51}$$

The two expressions share the same structure. Taking $\mathrm{Re}[q_j\,\cdot\,]$ and $\mathrm{Re}[\bar{q}_j\,\cdot\,]$, respectively,

$$\mathcal{S}_s^{(j)}(z) \;=\; \frac{\mathrm{Re}[q_j]\big(|E_f|^2 - |E_b|^2\big) \;-\; 2\,\mathrm{Im}[q_j]\,\mathrm{Im}\big(E_f E_b^*\big)}{\mathrm{Re}[q_0]}, \tag{52}$$

$$\mathcal{S}_p^{(j)}(z) \;=\; \frac{\mathrm{Re}[\bar{q}_j]\big(|E_f|^2 - |E_b|^2\big) \;-\; 2\,\mathrm{Im}[\bar{q}_j]\,\mathrm{Im}\big(E_f E_b^*\big)}{\mathrm{Re}[\bar{q}_0]}. \tag{53}$$

The first term is the net forward-minus-backward power and survives in a lossless layer; the second is the interference between counter-propagating waves and vanishes whenever the relevant wavenumber (q_j for s, $\bar{q}_j$ for p) is real.

Evaluating (46) at $z = z_{j-1}$ (left edge, $z - z_{j-1} = 0$) gives

$$E_f = \alpha_j^{(L)}, \quad E_b = \beta_j^{(L)}, \quad E_f E_b^* = \alpha_j^{(L)}\, \beta_j^{(L)\,*}, \tag{54}$$

and at $z = z_j$ (right edge), using (28),

$$E_f = \alpha_j^{(R)}, \quad E_b = \beta_j^{(R)}, \quad E_f E_b^* = \alpha_j^{(R)}\, \beta_j^{(R)\,*}. \tag{55}$$

The corresponding edge fluxes follow by substituting into (52)–(53):

$$\mathcal{S}_j^{(L)} \;=\; \mathcal{S}_{\mathrm{pol}}^{(j)}(z_{j-1}), \qquad \mathcal{S}_j^{(R)} \;=\; \mathcal{S}_{\mathrm{pol}}^{(j)}(z_j), \tag{56}$$

with $\mathrm{pol} \in \{s, p\}$ chosen consistently.

Applying energy conservation to the slab $z_{j-1} \le z \le z_j$ (with plane-wave geometry eliminating lateral fluxes) gives the absorbed fraction of incident power in layer j:

$$\boxed{A_j \;=\; \mathcal{S}_j^{(L)} \;-\; \mathcal{S}_j^{(R)}} \tag{57}$$

In numerical code, both edge amplitudes are available as partial products of the transfer-matrix chain, so (57) is evaluated directly via (47) and (48).

Since the Poynting flux is continuous across each interface, $\mathcal{S}_j^{(R)} = \mathcal{S}_{j+1}^{(L)}$. Summing (57) from $j = 1$ to N gives a telescoping result,

$$\sum_{j=1}^{N} A_j \;=\; \mathcal{S}_1^{(L)} \;-\; \mathcal{S}_N^{(R)} \;=\; \mathcal{S}_{\text{in}} \;-\; T, \tag{58}$$

where $\mathcal{S}_{\text{in}} \equiv \mathcal{S}_1^{(L)}$ is the flux immediately inside the first interface and $T = \mathcal{S}_N^{(R)}$ is the transmittance (the substrate carries no backward wave, $B_{N+1} = 0$). For a non-absorbing incident medium, $\mathcal{S}_{\text{in}} = 1 - R$ and (58) reduces to

$$R \;+\; T \;+\; \sum_{j=1}^{N} A_j \;=\; 1. \tag{59}$$

When the incident medium is itself absorbing, $\mathcal{S}_{\text{in}} \neq 1 - R$: interference between the incoming and reflected waves in front of the stack shifts the power-normalization baseline. This case is analyzed in detail by Byrnes [3, App. B]; equation (58) remains valid and is the most convenient invariant for code verification.

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